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Fire and Safety Engineering Department

"Studying Flammability Characteristics Of Bioplastic"

A Graduation Project Is Submitted To The Fire And Safety Engineering

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DEDICATION

To the great minds who paved the way before us—those upon whose shoulders we stand as we walk the path of knowledge and reason.

And to our families, especially **our parents**, whose unwavering support, prayers, and patience played the greatest role in guiding us to this milestone. To the engineers and doctors who have guided us with wisdom and patience throughout our journey, we extend our heartfelt appreciation.

We also express our deepest gratitude to our supervisor, **Dr. Diana Rbehat**, who spared no effort in encouraging us, inspiring us to give our best, and providing our team with invaluable insights and guidance that elevated our work at every stage.

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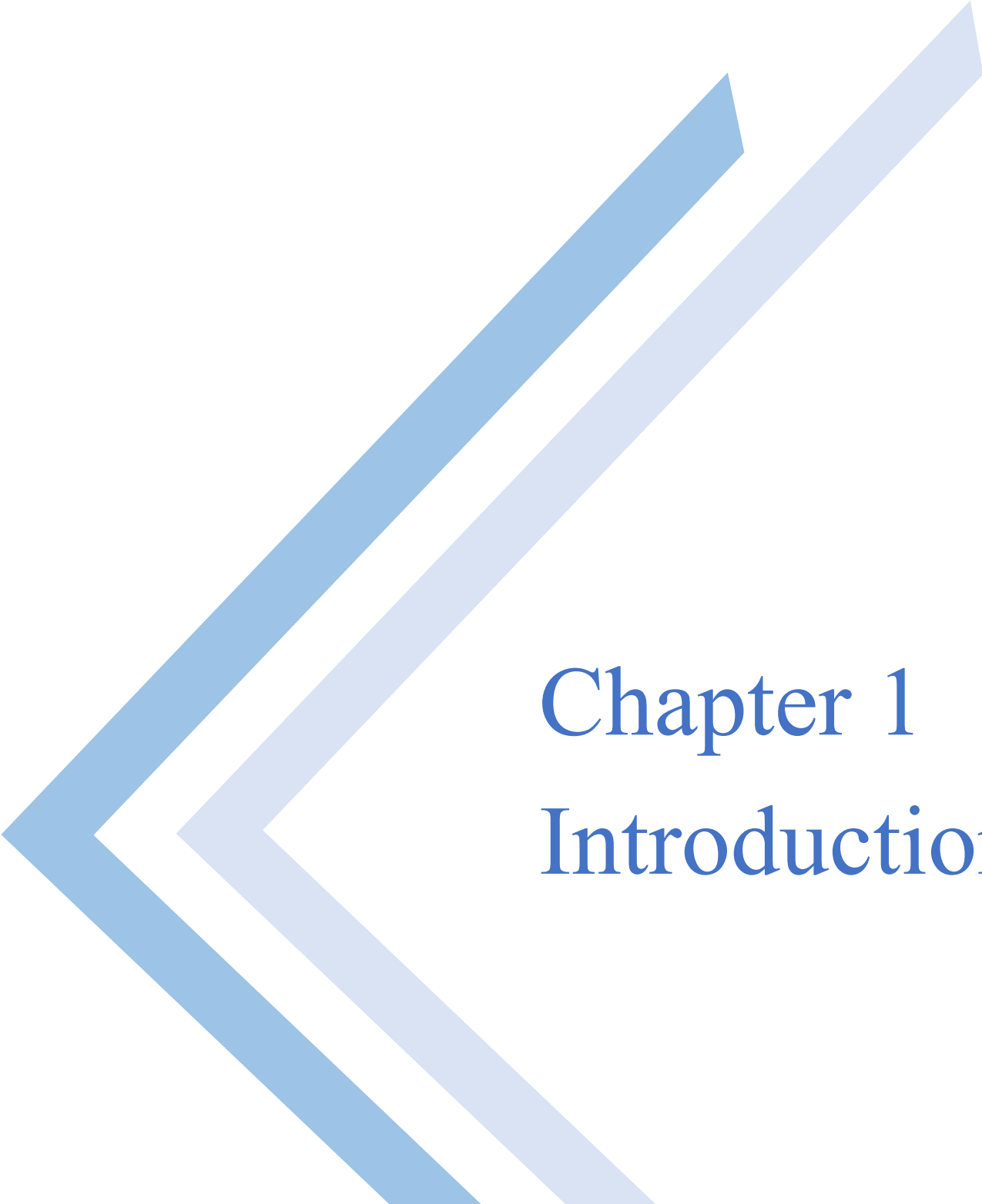
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Chapter 1

Introduction

1.1 Genaral Introduction

In recent decades, plastic materials have become deeply integrated into modern construction, interior finishing, industrial applications, and consumer products. Their widespread adoption is attributed to properties such as durability, low density, ease of manufacturing, and economic efficiency [1].

However—despite these advantages—plastics represent one of the **most critical challenges in fire safety engineering**, particularly due to their behavior under elevated temperatures and flame exposure.

When subjected to heat, conventional plastics undergo rapid **thermal decomposition**, releasing significant amounts of heat and producing dense smoke rich in toxic combustion gases, chiefly carbon monoxide [2]. This smoke not only accelerates fire growth but also leads to **drastic visibility reduction**, making evacuation more difficult and increasing the likelihood of injuries and fatalities. Numerous studies highlight that smoke inhalation remains the leading cause of death in modern fires involving polymeric materials [3].

Beyond fire-related hazards, traditional plastics pose a considerable **environmental challenge**, as they may require centuries to degrade and accumulate extensively in soil and marine environments [4]. Their manufacturing relies heavily on non-renewable petrochemical resources, contributing to increased carbon emissions and global pollution.

These concerns have motivated the search for materials that provide a balance between **performance, safety, and sustainability**. Among these alternatives, **bio-plastic** has gained attention due to its origin from renewable, biodegradable sources [5]. Bio-plastics not only reduce long-term environmental impact but may also exhibit **distinct fire behavior** arising from their different chemical structure, influencing characteristics such as ignitability, heat release rate, and smoke production [6].

Accordingly, this project aims to provide a systematic and scientifically grounded evaluation of the fire flammability characteristics of bio-plastic materials, with the objective of analyzing their

behavior under fire exposure and assessing their suitability from a fire safety engineering perspective.

1.2 Aim and Objectives

This project aims to examine the fire flammability of bio-plastic materials when exposed to fire and to investigate the possibility of modifying their composition in order to improve fire resistance, with the aim of considering bio-plastics as a potential alternative to conventional plastics.

Titles:

This aim will be achieved through the following:

1.2.1 Selecting suitable bio-plastic samples for the purpose of the study.

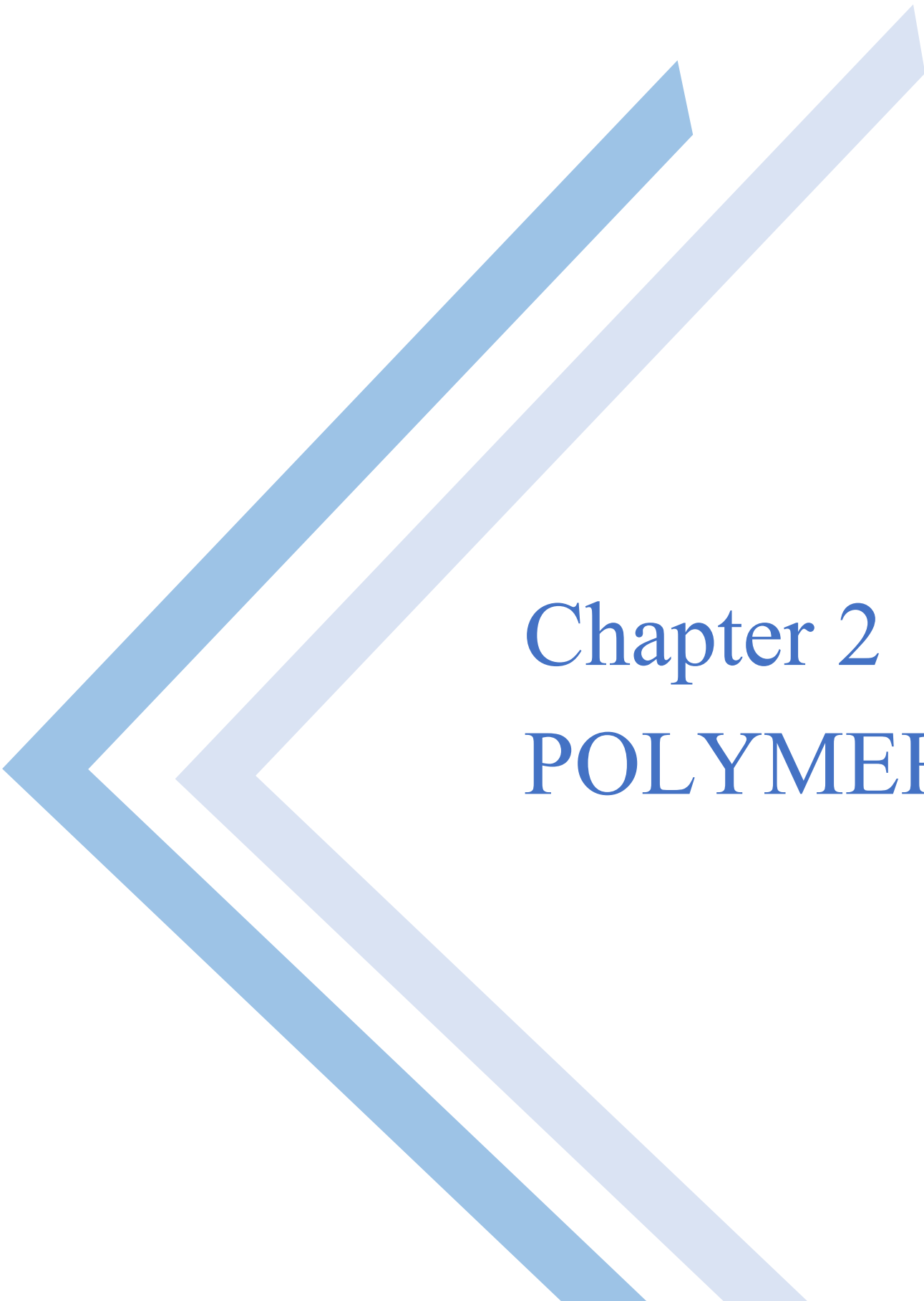
1.2.2 Preparing bio-plastic materials and applying compositional modifications to enhance their fire resistance.

1.2.3 Conducting fire-related tests to evaluate the flammability behavior of bio-plastic materials before and after modification.

1.2.4 Comparing the fire response of modified bio-plastics with that of conventional plastics.

1.3 Literature Review

(نشوف المقالات السابقة من الجامعات وغيرها ونلخصهن)



Chapter 2

POLYMERS

2.1 Introduction to Polymers

Polymers constitute a broad class of materials composed of long-chain macromolecules formed through the repetitive linkage of smaller molecular units known as monomers. These macromolecular chains are interconnected by covalent bonds, resulting in materials that exhibit a wide range of physical, chemical, and thermal properties. The ability to tailor polymer structure at the molecular level enables the design of materials suitable for diverse engineering applications.

In modern engineering practice, polymers have become indispensable materials in sectors such as construction, transportation, electrical insulation, packaging, and consumer products. Their widespread utilization is primarily attributed to their low density, ease of processing, resistance to corrosion, and economic advantages compared to conventional materials such as metals and ceramics.

Despite these advantages, polymers display distinct behavior when exposed to elevated temperatures and fire conditions. Owing to their organic chemical structure, most polymers are combustible and can significantly contribute to fire development through heat release, smoke generation, and emission of toxic gases. Consequently, understanding the thermal and fire behavior of polymers is essential in fire safety engineering and provides the theoretical foundation for the experimental investigations presented in this research [7].

2.2 Types of Polymers

Polymers can be classified according to their origin, molecular structure, and thermal behavior. A general classification of polymer materials is presented in **Figure (2.1)**.

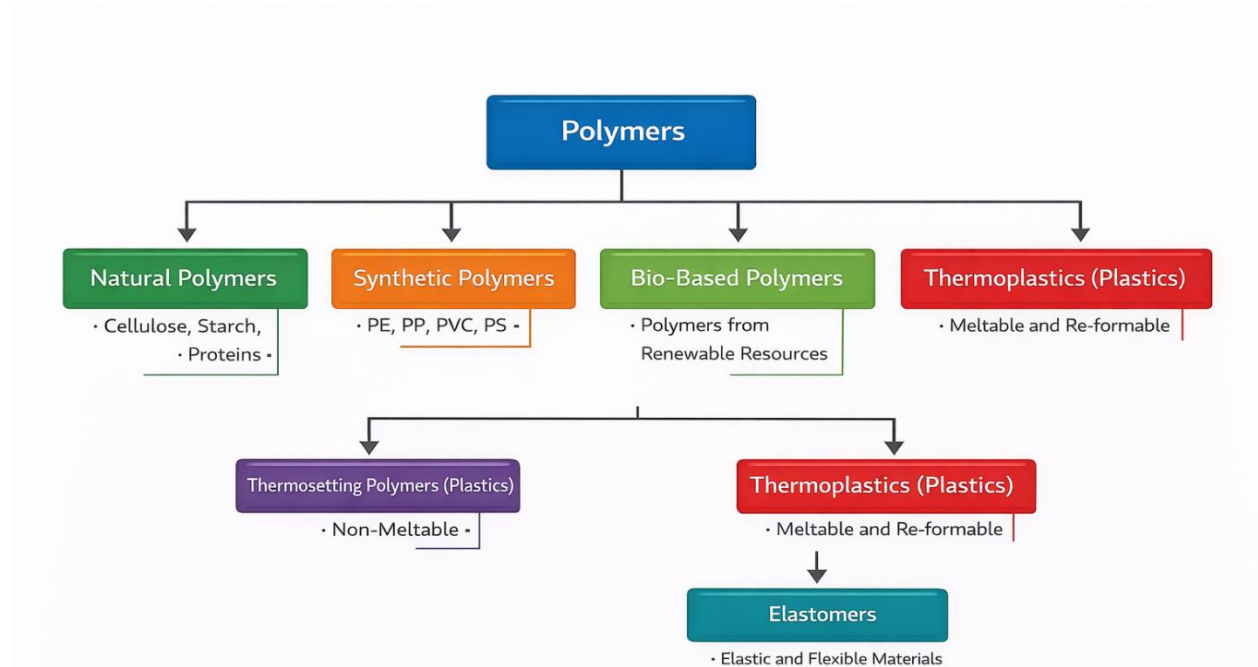


Figure 2.1: Classification of Polymers

2.2.1 Natural Polymers

Natural polymers are derived from biological sources and occur naturally in living organisms. Typical examples include cellulose, starch, proteins, and natural rubber. These materials are generally biodegradable and environmentally benign; however, their limited thermal stability and weak resistance to fire restrict their application in fire-sensitive environments [8].

2.2.2 Synthetic Polymers

Synthetic polymers are produced through industrial polymerization processes, primarily from petroleum-based raw materials. Common examples include polyethylene (PE), polypropylene (PP), polystyrene (PS), and polyvinyl chloride (PVC). These polymers dominate modern plastic applications due to their durability, versatility, and ease of large-scale production [⁹].

2.2.3 Bio-Based Polymers

Bio-based polymers are partially or fully derived from renewable biological resources such as plants or microorganisms. While some bio-based polymers are biodegradable, others may exhibit chemical structures similar to conventional plastics and therefore do not readily degrade. Their principal advantage lies in reducing dependence on fossil resources rather than guaranteeing biodegradability [¹⁰].

2.2.4 Thermoplastics (Plastics)

Thermoplastics represent the most extensively used class of polymer materials and constitute the primary focus of this study. These polymers soften upon heating and solidify upon cooling, allowing repeated reshaping and recycling. Common thermoplastics include PVC, PE, PP, and polyethylene terephthalate (PET).

From a fire safety perspective, thermoplastics present several critical challenges. When exposed to heat, thermoplastics initially soften and melt, followed by **thermal decomposition (pyrolysis)**. During pyrolysis, polymer chains break down and release flammable volatile compounds. Once these gases reach their ignition temperature, combustion occurs, resulting in rapid fire growth and high heat release rates.

Furthermore, thermoplastics are known to generate dense smoke and toxic combustion products. For instance, PVC releases hydrogen chloride (HCl) gas during combustion, which poses severe health hazards and contributes to corrosion of building components. These characteristics render conventional plastics a significant concern in fire safety engineering [¹¹].

From an environmental standpoint, conventional plastics exhibit extremely slow degradation rates in soil and marine environments, often requiring decades or even centuries to decompose. This persistence leads to long-term environmental pollution and accumulation in ecosystems [12].

2.2.5 Thermosetting Polymers

Thermosetting polymers undergo irreversible chemical cross-linking during the curing process, forming rigid three-dimensional molecular networks. Once cured, thermosets do not melt upon reheating but instead undergo thermal decomposition. Although thermosetting polymers generally exhibit superior thermal stability compared to thermoplastics, their brittleness and lack of recyclability limit their practical applications [13].

2.2.6 Elastomers

Elastomers are polymers characterized by high elasticity and flexibility, such as natural rubber and synthetic elastomers. While these materials can effectively absorb mechanical energy, they are typically combustible and may produce hazardous smoke and gases when exposed to fire conditions [14].

2.3 Fire Behavior of Polymers

The fire behavior of polymers is governed by their chemical composition, molecular structure, and thermal properties. When exposed to an external heat source, polymers do not ignite immediately; instead, they undergo a sequence of thermally induced processes.

Initially, the polymer absorbs heat, resulting in a rise in temperature and subsequent softening or melting. Continued heating leads to **pyrolysis**, during which polymer chains decompose and release flammable gaseous products. These gases mix with ambient oxygen and ignite once favorable temperature and oxygen concentration conditions are achieved [15].

Polymer combustion is typically associated with **high heat release rates (HRR)**, which significantly accelerate fire growth and intensify fire severity. In addition, polymer fires produce

large quantities of dense smoke and toxic gases such as carbon monoxide (CO), hydrogen cyanide (HCN), and acidic compounds, depending on polymer type. These by-products reduce visibility and pose serious risks to occupants through inhalation toxicity [16].

Understanding these fire-related characteristics is essential for evaluating polymer performance and selecting appropriate experimental testing methods, such as the Cone Calorimeter, Thermogravimetric Analysis (TGA), Differential Scanning Calorimetry (DSC), and thermal conductivity measurements discussed in **Chapter 3**.

2.4 Bio-Plastics

Bio-plastics are polymeric materials manufactured partially or entirely from renewable biological resources rather than conventional petroleum-based feedstocks. Typical sources include corn starch, sugarcane, cellulose, vegetable oils, and materials produced through biological processes involving microorganisms [17, 18]. Depending on their chemical composition and production route, bio-plastics may be classified as bio-based, biodegradable, or both.

Bio-plastics are produced through chemical or biological processes that convert natural raw materials into moldable polymers suitable for conventional plastic manufacturing techniques. Common examples include **polylactic acid (PLA)** and **poly hydroxy alkanoates (PHA)**, which can be processed using extrusion, injection molding, and thermoforming methods similar to those employed for traditional plastics [19]. This compatibility with existing processing technologies has facilitated the industrial adoption of bio-plastics and supports their integration within sustainability and circular economy frameworks [20].

Bio-plastics are currently utilized in a wide range of applications, including packaging, medical products, consumer goods, and selected engineering components. Continuous research and development efforts aim to enhance their mechanical properties, thermal stability, and durability in order to broaden their applicability in more demanding environments [21].

From a fire safety standpoint, bio-plastics exhibit thermal and combustion behavior that differs from conventional plastics due to their oxygen-rich molecular structures. Upon exposure to elevated temperatures, bio-plastics undergo thermal degradation and pyrolysis, releasing combustible gases in a manner similar to petroleum-based plastics. However, variations may be observed in decomposition temperature, heat release rate, smoke generation, and char formation, depending on material composition.

Certain bio-plastics demonstrate an increased tendency to form char during combustion, which may act as a protective thermal barrier that slows heat transfer and reduces material degradation. Nevertheless, many bio-plastics still exhibit limited inherent fire resistance, including relatively low ignition temperatures and significant smoke production, thereby restricting their use in fire-sensitive building applications unless further material modification is applied [22].

Accordingly, a central research question arises as to whether bio-plastics can serve as safer alternatives to conventional plastics when exposed to fire. Addressing this question requires a systematic evaluation of key fire-related parameters, including **heat release rate, mass loss rate, smoke generation, and thermal conductivity**, which form the basis of the experimental investigations presented in **Chapter 3**.

2.5 Comparison Between Conventional Plastics and Bio-Plastics

Conventional plastics and bio-plastics differ significantly in terms of origin, environmental impact, and fire behavior. Conventional plastics provide excellent durability and mechanical performance; however, they are associated with high flammability, dense smoke production, toxic gas emissions, and extremely slow environmental degradation.

In contrast, bio-plastics offer environmental advantages through the use of renewable resources and, in some cases, improved biodegradability. Despite these benefits, bio-plastics often exhibit inferior fire performance compared to conventional plastics unless appropriate modifications are implemented.

Table (2.1) presents a comparative summary of key differences between conventional plastics and bio-plastics.

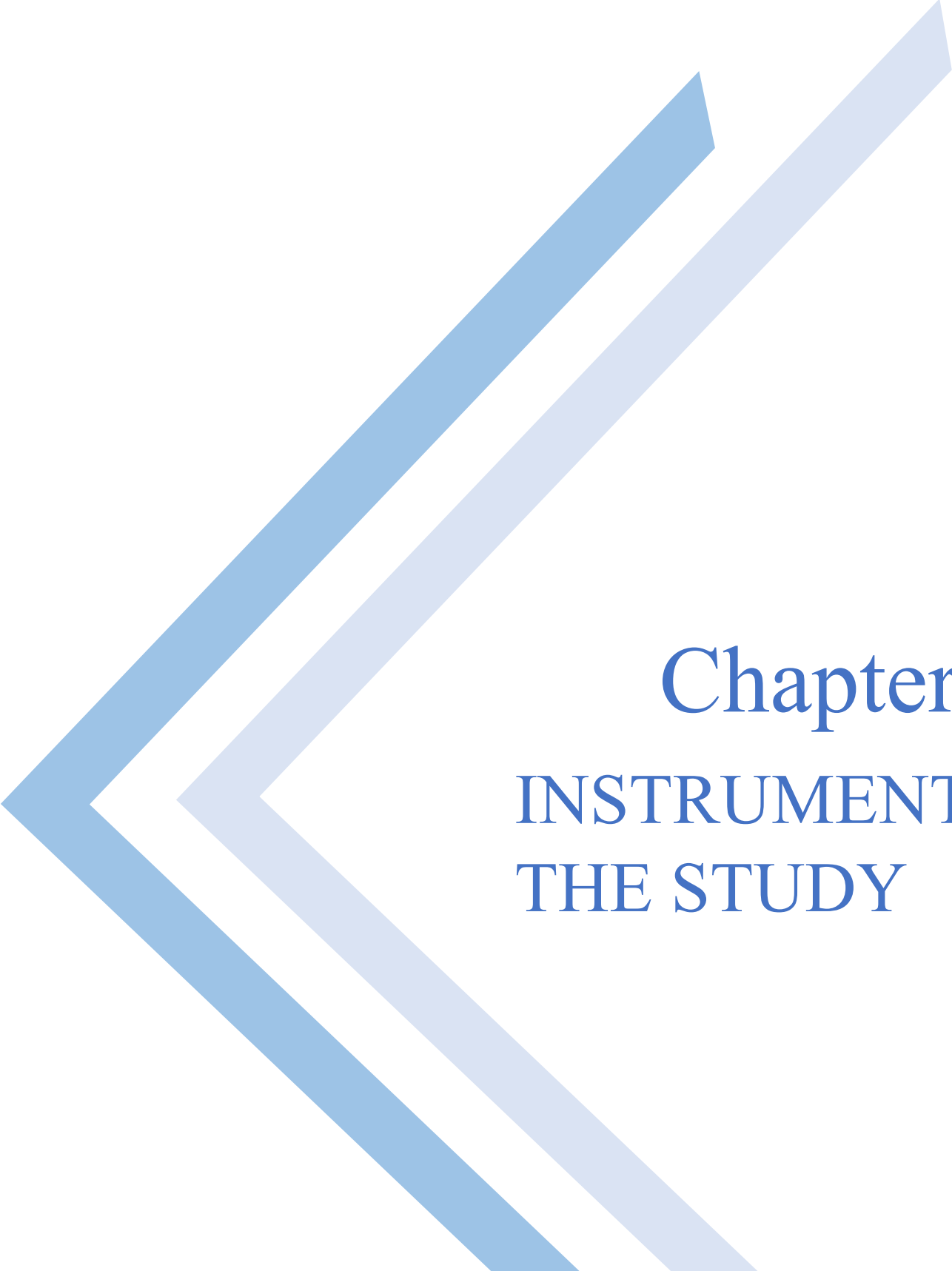
Table (2.1): Comparison Between Conventional Plastics and Bio-Plastics

Aspect	Conventional Plastics	Bio-Plastics
Raw material source	Petroleum-based	Renewable biological resources
Environmental impact	High, non-biodegradable	Reduced, partially biodegradable
Thermal stability	Moderate to high	Generally lower
Fire behavior	High flammability, high HRR	Variable, often lower HRR
Smoke and toxic gases	Dense and toxic	Depends on composition
Degradation in soil	Very slow (decades–centuries)	Faster, depending on type

2.6 Chapter Summary

This chapter presented a comprehensive overview of polymeric materials, including their classification, structural characteristics, and behavior under fire exposure. Particular emphasis was placed on conventional plastics due to their widespread use and associated fire and environmental challenges. Bio-plastics were discussed as promising alternatives, highlighting both their advantages and current limitations in fire-related applications.

The concepts developed in this chapter provide the theoretical foundation for **Chapter 3**, which introduces the experimental instruments used to evaluate thermal stability, fire behavior, and thermal conductivity of polymeric materials. These standardized testing techniques enable a systematic comparison between conventional plastics and bio-plastics under controlled laboratory conditions.



Chapter 3

INSTRUMENTS OF THE STUDY

3.1 Cone Calorimeter

3.1.1 Introduction

The Cone Calorimeter is a standard fire-testing device based on the principle of oxygen consumption during combustion. It is widely used in fire research laboratories to evaluate the fire behavior of materials under controlled conditions. Due to its reliability and repeatability, the cone calorimeter is considered the most significant bench-scale instrument in fire testing and is commonly used to provide input data for fire modeling and simulation studies.

The device exposes the surface of a test specimen to a predefined external radiant heat flux. Once ignition occurs, the specimen burns in excess air, allowing several fire-related parameters to be measured. These parameters include time to ignition, heat release rate, mass loss rate, smoke production, and gas emissions, which together describe the combustion behavior of the material.

The cone calorimeter operates according to internationally recognized standards, namely ISO 5660-1 and ASTM E1354, ensuring consistency and reproducibility of test results [²³, ²⁴].

3.1.2 General Description

The cone calorimeter consists of several integrated components that work together to control, measure, and record the combustion process. These components include the cone heater, specimen holder, load cell, exhaust system, oxygen analyzer, and data acquisition system. A schematic representation of these components is shown in **Figure (3.1)**.

The device allows for precise control of heat flux levels, enabling systematic evaluation of material behavior under different fire exposure conditions.



Figure (3.1)

3.1.3 Parameters Measured

The cone calorimeter provides a comprehensive set of fire performance parameters, including:

- Time to ignition.
- **Heat Release Rate (HRR).**
- **Mass Loss Rate (MLR).**
- Total heat released.
- Smoke production.
- Toxic gas yields (depending on analyzer configuration).

In the present research, particular emphasis is placed on **heat release rate** and **mass loss rate**, as these parameters are critical indicators of fire intensity and material flammability.

3.2 Thermogravimetric Analysis (TGA)

3.2.1 Introduction

Thermogravimetric Analysis (TGA) is a thermal analysis technique used to evaluate the mass change of a material as a function of temperature or time under a controlled atmosphere. TGA is widely applied in polymer science and fire research to assess **thermal stability**, **decomposition behavior**, and **residual char formation**, all of which are relevant to fire performance.

3.2.2 Principle of Operation

In a TGA test, a small sample is placed on a high-precision microbalance inside a temperature-controlled furnace. The sample is subjected to a predefined heating program while exposed to an inert or oxidative atmosphere. As temperature increases, physical or chemical changes such as evaporation, decomposition, or oxidation occur, leading to measurable mass loss.

The instrument continuously records mass changes, allowing identification of degradation stages and decomposition mechanisms [25].

Figure (3.2) illustrates the Thermogravimetric Analysis (TGA) device used for evaluating the thermal stability of the tested materials.

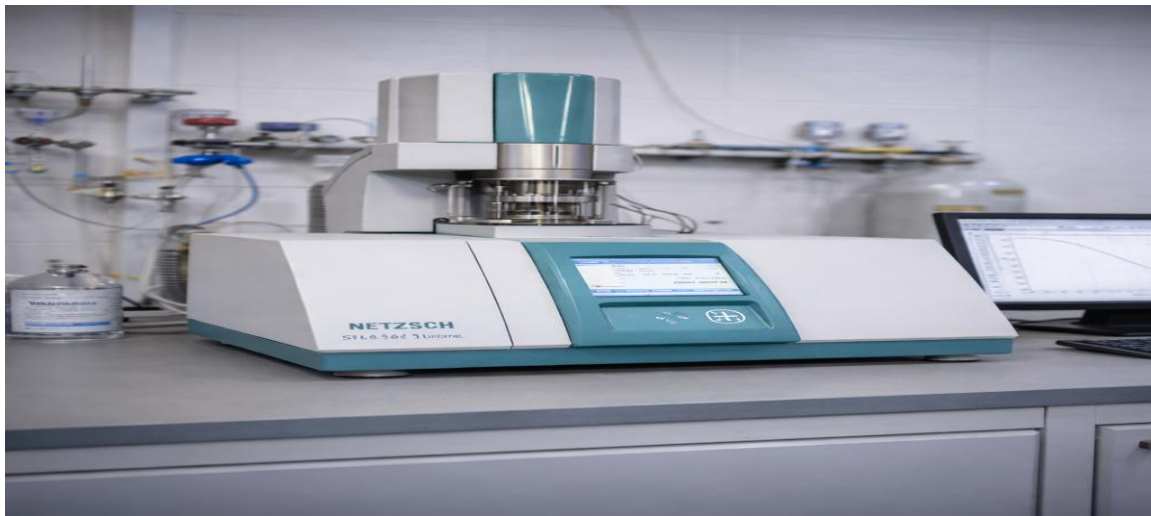


Figure (3.2)

3.2.3 Parameters Measured

TGA provides the following key outputs:

- Onset decomposition temperature.
- Mass loss stages and rates.
- Residual mass (char yield or ash content).
- Derivative thermogravimetric (DTG) curve.

These parameters are essential for evaluating the effectiveness of material modifications aimed at improving fire resistance.

3.3 Differential Scanning Calorimetry (DSC)

3.3.1 Introduction

Differential Scanning Calorimetry (DSC) is a thermal analysis technique that measures heat flow associated with thermal transitions in materials as a function of temperature. DSC is commonly used to characterize polymers by identifying transitions such as glass transition, melting, and crystallization, which influence material behavior under thermal and fire exposure.

3.3.2 Principle of Operation

In DSC analysis, both the sample and a reference material are subjected to the same controlled temperature program. The instrument measures the difference in heat flow required to maintain both at the same temperature. Thermal transitions within the sample appear as endothermic or exothermic peaks on the DSC thermogram.

This method provides insight into structural and thermal changes occurring within the polymer matrix [26]. **Figure (3.3)** illustrates the Differential Scanning Calorimetry (DSC) device used to examine the thermal behavior of the tested materials.



Figure (3.3)

3.3.3 Parameters Measured

DSC allows determination of:

- Glass transition temperature (T_g).
- Melting temperature (T_m).
- Crystallization temperature (T_c).
- Enthalpy changes (ΔH).
- Degree of crystallinity (when applicable).

These parameters help assess how compositional modifications affect thermal behavior and potentially improve fire resistance.

3.4 Thermal Conductivity Measurement (TCA 300-DTX)

3.4.1 Introduction

Thermal conductivity is a fundamental property that describes a material's ability to conduct heat and is expressed in $\text{W}/(\text{m}\cdot\text{K})$. Materials with low thermal conductivity are particularly valuable in fire safety and building applications due to their enhanced insulation performance.

In this study, thermal conductivity measurements were conducted using the **TCA 300-DTX** device manufactured by **Taurus Instruments GmbH**.

3.4.2 Principle of Operation

The TCA 300-DTX operates based on the **heat flow meter method**, where a specimen is placed between two plates maintained at different temperatures. A steady-state temperature gradient is established, and the heat flux through the specimen is measured. Thermal conductivity is calculated using the measured heat flow, specimen geometry, and temperature difference.

The measurement procedure complies with **ISO 8301** and **ASTM C518**, ensuring accuracy and reproducibility of results [^{27, 28}].

Figure (3.4) illustrates the TCA 300-DTX thermal conductivity measurement device used to determine the thermal conductivity of the tested materials.



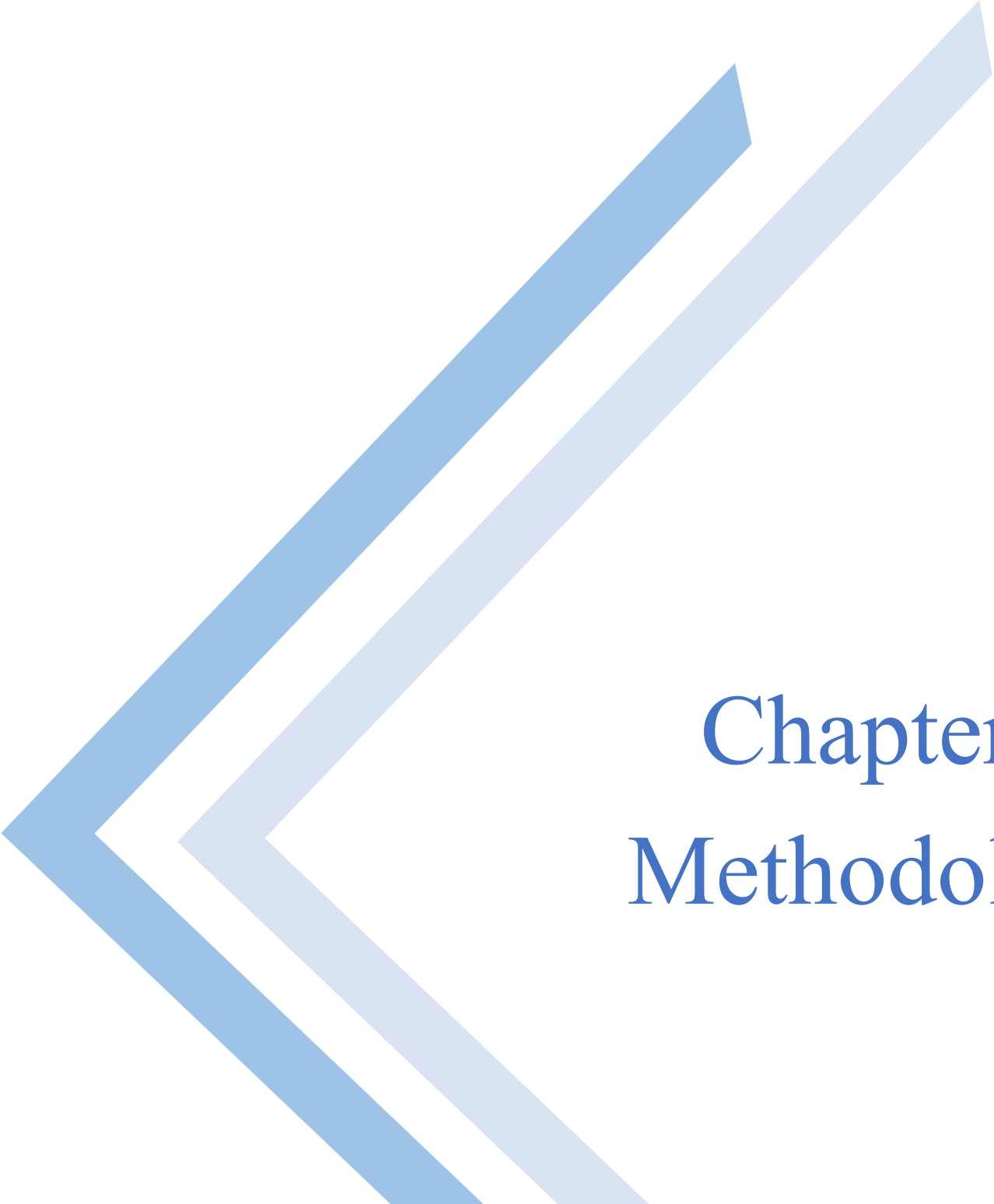
Figure (3.2)

3.4.3 Parameters Measured

The primary parameter obtained from the TCA 300-DTX test is:

- **Thermal conductivity (k)**

This parameter complements fire testing by providing information on the insulation performance of the materials under thermal exposure.



Chapter 4

Methodology

Abstract

This research aims to evaluate the **thermal performance of bio-plastic during fire exposure**, within the broader context of public safety studies focused on minimizing risks and protecting both human life and the environment. Experimental tests were carried out using the **TCA DTX 300** and **Cone Calorimeter** devices to measure thermal conductivity, heat release rate (HRR), mass loss, carbon monoxide (CO) emissions, smoke density, and the temperature of smoke generated during combustion.

The samples were prepared with dimensions of **100 mm × 100 mm**, a thickness of **** mm, and a weight of **** grams. These dimensions were selected to ensure accuracy and reliability across the various fire-performance tests. (نكتب ملخص بسيط)
عن نتائج العينات

The findings contribute to understanding the material's ability to **withstand fire** and its thermal response when exposed to elevated temperatures, thus supporting the assessment of its suitability for use in applications requiring high safety standards. Furthermore, the study highlights the importance of selecting materials that help reduce hazards, limit smoke and toxic gas production, and enhance protection for individuals and the environment in potential fire scenarios.

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